

Transcriptome-Driven Drug Repositioning to Reverse T Cell Exhaustion Using Large-Scale Perturbation Signatures

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Abstract

T cell exhaustion is a dysfunctional differentiation state of CD8⁺ T cells that arises in chronic antigen exposure settings such as cancer and persistent viral infection. Although immune checkpoint inhibitors partially restore T cell activity in subsets of patients, their limited response rates and toxicity profiles motivate the search for complementary strategies. Here, we present an AI-assisted, transcriptome-driven drug repositioning framework to identify approved small-molecule compounds capable of reversing exhausted T cell states. By integrating bulk and single-cell RNA sequencing data with large-scale perturbation signatures from the LINCS L1000 resource, we construct a robust exhaustion-associated gene expression signature and computationally screen drugs that induce inverse transcriptional programs. The framework produces a prioritized list of candidate repositioning drugs and provides mechanistic hypotheses connecting these compounds to core regulatory pathways of T cell exhaustion. This study demonstrates how public omics data and AI-driven analysis can accelerate hypothesis generation for translational immunotherapy research.

1 Introduction

T cell exhaustion represents a major barrier to durable immune-mediated control of cancer and chronic infection. Under sustained antigen stimulation, CD8⁺ T cells progressively lose effector functions, including cytokine production, cytotoxicity, and proliferative capacity, while upregulating inhibitory receptors such as PD-1, TIM-3, LAG-3, and TIGIT. At the molecular level, exhaustion is now understood as a stable transcriptional and epigenetic program rather than a transient functional impairment.

Immune checkpoint blockade has revolutionized cancer therapy; however, only a fraction of patients achieve durable responses, and immune-related adverse events remain a significant concern. These limitations motivate the exploration of alternative or complementary approaches capable of modulating exhausted T cell states. Drug repositioning offers a promising strategy by identifying new immunological applications for compounds with established clinical safety profiles.

Recent advances in transcriptomics and public perturbation databases enable systematic, data-driven drug repositioning. By comparing disease-associated gene expression signatures with drug-induced transcriptional responses, it is possible to identify compounds that may reverse pathological cellular states. In this study, we apply this paradigm to T cell exhaustion and propose a scalable computational framework for identifying candidate drugs that induce inverse exhaustion-associated transcriptional programs.

2 Background and Related Work

Extensive transcriptomic and epigenomic studies have revealed that exhausted T cells occupy a distinct cellular state governed by key transcription factors, including TOX and the NR4A family. Single-cell RNA sequencing further demonstrated that exhaustion comprises heterogeneous subpopulations, such as progenitor exhausted and terminally exhausted T cells, with different therapeutic susceptibilities.

Parallel to these biological insights, computational drug repositioning approaches such as the Connectivity Map and the LINCS project have provided large-scale perturbation datasets linking drugs, genes, and cellular states

through transcriptional signatures. These resources have been successfully applied to cancer, neurological disorders, and inflammatory diseases. However, systematic application of transcriptome-based drug repositioning to T cell exhaustion remains limited, motivating the present study.

3 Objectives and Research Questions

The primary objective of this work is to identify approved drugs capable of reversing T cell exhaustion at the transcriptomic level. The study addresses three core research questions: (1) Can a standardized and reproducible gene expression signature of T cell exhaustion be defined across bulk and single-cell datasets? (2) Which approved drugs induce transcriptional programs inversely correlated with this signature? (3) Through which molecular pathways might these drugs modulate exhaustion-associated regulatory networks?

4 Methods

4.1 Data Collection

Bulk RNA-seq and single-cell RNA-seq datasets of exhausted and functional CD8⁺ T cells were collected from public repositories, including GEO and SRA. Datasets were selected based on clear experimental annotations, sufficient sample sizes, and relevance to cancer or chronic infection contexts.

4.2 Signature Construction

Differential expression analysis was performed between exhausted and functional T cell populations. Genes meeting predefined statistical thresholds were selected to define an exhaustion-associated signature. Pathway enrichment and co-expression analyses were applied to capture system-level features.

4.3 Drug Signature Matching

Drug-induced transcriptional signatures were obtained from the LINCS L1000 dataset. Similarity between drug signatures and the exhaustion signature was quantified using multiple metrics, including Pearson and Spearman correlations and cosine similarity. Drugs consistently exhibiting inverse correlations were prioritized.

5 Results

The exhaustion signature captured canonical inhibitory receptors, transcriptional regulators, and metabolic pathways associated with dysfunctional T cell states. Inverse signature matching identified multiple approved drugs exhibiting strong negative correlations with the exhaustion signature across multiple metrics and experimental conditions.

6 Discussion

The results demonstrate the feasibility of transcriptome-driven drug repositioning for immunotherapy. By focusing on inverse transcriptional programs rather than individual targets, the framework captures system-level regulatory effects. The identified candidates generate testable hypotheses for restoring T cell function and complementing existing immunotherapeutic strategies.

7 Limitations

This study relies on in silico analyses of public datasets and does not include experimental validation. Context-specific effects, dataset heterogeneity, and cell line differences in LINCS represent important limitations that should be addressed in future work.

8 Conclusion

We present an AI-assisted computational framework for identifying drug repositioning candidates capable of reversing T cell exhaustion. By integrating transcriptomic signatures with large-scale perturbation data, this approach provides a scalable foundation for hypothesis generation and translational immunotherapy research.

References

References omitted for brevity.